

SHAHRUD, Iran, Islamic Republic of

WMO: 407390

Lat: 36.3801N

Lon: 54.9292E

Elev: 4426

StdP: 12.49

Time Zone: 3.50 (IRN)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.6	24.6	4.3	8.1	35.4	8.2	9.8	35.7	18.2	36.1	14.8	40.2	2.1	330	0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	96.7	63.8	94.2	63.1	91.8	62.3	67.2	89.2	65.9	87.7	64.8	86.2	6.9	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.3	89.0	79.5	57.7	84.0	78.5	56.4	79.9	77.9	34.4	89.2	33.2	87.9	32.3	86.4	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.4	16.9	14.3	17.7	101.6	4.6	1.8	14.4	102.9	11.8	103.9	9.2	104.9	5.9	106.2
			15.4	69.8	4.2	2.1	12.4	71.3	10.0	72.5	7.6	73.6	4.6	75.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	59.8	36.1	40.2	49.2	59.4	69.6	78.0	82.0	80.0	73.2	62.3	48.0	38.5
	DBStd	17.35	5.63	6.73	7.94	7.50	5.71	5.39	5.01	4.39	5.23	6.35	7.24	5.36
	HDD50	1309	430	278	110	15	0	0	0	0	1	119	356	
	HDD65	3759	895	693	491	194	21	0	0	5	129	511	820	
	CDD50	4896	0	5	86	298	608	839	991	932	695	383	58	1
	CDD65	1874	0	0	3	27	165	390	526	467	250	46	0	0
	CDH74	18603	0	0	19	207	1297	3967	5973	4804	2039	297	0	0
	CDH80	8074	0	0	2	35	372	1762	2957	2192	713	41	0	0
Wind	WSAvg	5.0	2.8	3.9	4.7	5.3	5.9	7.4	8.1	7.1	5.1	3.9	3.1	2.5
Precipitation	PrecAvg	6.2	0.6	0.9	1.1	1.0	0.7	0.3	0.2	0.1	0.2	0.2	0.4	0.6
	PrecMax	8.3	1.8	3.6	2.7	3.6	2.1	0.9	0.8	0.5	1.1	1.1	1.3	1.5
	PrecMin	2.6	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	1.6	0.4	0.7	0.6	0.8	0.6	0.2	0.2	0.1	0.2	0.3	0.4	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.6	62.6	76.1	84.2	90.6	97.2	100.7	98.6	93.0	84.3	70.8	57.8
		MCWB	41.7	45.0	51.5	55.4	59.4	62.7	65.1	62.9	60.2	59.1	51.4	44.0
	2%	DB	50.6	57.1	69.7	79.9	87.1	94.5	97.6	95.3	89.8	80.3	66.4	52.9
		MCWB	39.9	42.4	48.3	54.4	58.4	61.8	64.8	63.1	60.1	56.3	49.2	41.9
	5%	DB	47.8	53.8	65.6	75.9	84.2	91.8	95.2	92.7	87.1	77.4	62.8	50.0
		MCWB	38.3	41.0	46.9	52.8	57.9	61.2	64.4	62.7	59.4	55.0	47.6	40.4
	10%	DB	45.0	51.0	62.0	72.3	81.2	89.1	92.3	90.1	84.3	74.1	59.4	47.5
		MCWB	36.9	39.7	45.5	51.8	57.2	60.6	63.6	62.1	59.0	53.7	46.0	39.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.9	47.3	53.0	57.6	62.8	67.3	70.1	68.3	65.6	61.3	53.1	45.4
		MCDB	51.2	56.6	70.0	77.0	82.0	89.9	91.7	88.9	85.1	80.2	67.0	54.1
	2%	WB	41.2	44.4	50.6	55.8	60.5	65.2	67.8	66.6	63.8	58.4	50.7	43.4
		MCDB	47.7	53.9	64.7	74.0	81.2	87.6	90.5	87.2	83.1	75.9	63.1	50.8
	5%	WB	39.2	42.3	48.6	54.4	59.3	63.6	66.4	65.5	62.2	56.4	48.8	41.8
		MCDB	46.8	51.8	62.7	71.5	79.7	85.9	89.3	86.0	80.7	73.2	60.6	48.5
	10%	WB	37.5	40.6	46.7	53.1	58.1	62.1	65.2	64.5	60.9	54.9	47.0	40.1
		MCDB	44.5	49.8	60.2	69.2	78.1	84.1	87.6	84.7	79.1	71.0	57.9	46.3
Mean Daily Temperature Range	5% DB	MDBR	13.9	15.9	17.9	18.7	19.2	19.7	18.6	19.2	20.4	19.4	16.3	13.6
		MCDBR	16.6	18.6	21.2	21.5	21.7	21.5	20.6	21.3	22.4	21.9	18.9	16.7
		MCWBR	9.5	9.5	9.1	8.1	7.5	7.2	6.8	7.2	8.3	9.0	8.9	9.6
	5% WB	MCDBR	15.4	16.8	18.6	19.2	19.9	20.0	19.5	19.5	19.9	20.0	17.0	14.6
		MCWBR	9.1	9.4	9.3	8.2	7.6	7.9	7.2	7.4	8.2	8.9	8.4	8.8
Clear-Sky Solar Irradiance	taub	0.315	0.351	0.412	0.469	0.484	0.476	0.500	0.451	0.417	0.404	0.349	0.312	
	taud	2.319	2.203	2.019	1.915	1.913	1.955	1.909	2.041	2.107	2.111	2.248	2.340	
	Ebn at Noon	276	278	268	259	257	258	251	261	264	254	259	269	
	Edh at Noon	30	39	51	60	61	59	61	52	46	42	33	28	
All-Sky Solar Radiation	RadAvg	823	1083	1452	1791	2133	2458	2383	2225	1840	1372	957	737	
	RadStd	51	93	96	140	82	90	81	114	61	56	58	72	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)		+0.65	N/A	N/A	+1.60	+0.95	+1.04	N/A	N/A	N/A	N/A

Nomenclature: See separate page